

GOVT 6019: Homework 4 — Expectation

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Part A — Review: naming a distribution

Question 1: Name the distribution

For each scenario, name the distribution the random variable follows and give its parameters, with one sentence of justification.

- i. A state statute requires that the order of the 7 candidates on the ballot be randomized, with every ordering equally likely. X is the ballot position assigned to one named candidate.
- ii. One voter is sampled at random from a district. $X = 1$ if they support the incumbent and $X = 0$ otherwise.
- iii. Fifteen voters are sampled independently at random from a district in which 40 percent of voters support the incumbent. X is the number of sampled voters who support the incumbent.
- iv. Motions of no confidence are tabled in a parliament at an average rate of 1.5 per year. X is the number tabled in a given year.

Then compute the following.

- a. For (i): the probability that the named candidate is assigned the top position, and the probability that they are assigned one of the top two.
- b. For (iii) with four sampled voters rather than fifteen: list every sequence of four responses containing exactly two supporters, count them, and confirm your count against $\binom{4}{2}$. Then compute $\mathbb{P}(X = 2)$ by hand.
- c. For (iii) as stated, with fifteen sampled voters: compute $\mathbb{P}(X = 6)$ and $\mathbb{P}(X \leq 5)$ in R.

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# dbinom() gives the PMF; pbinom() gives the CDF
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Part B — Conditioning and pooled comparisons

Question 2: Two ways to reach the hospital

A regional trauma registry records every patient transported to hospital after a road accident, the mode of transport, the severity of the accident, and whether the patient died. Accidents are classified as serious or less serious at the scene, before transport is chosen. (The counts are stipulated for the exercise.)

Deaths / patients	Serious	Less serious	All
Helicopter	48/100	16/100	64/200
Ambulance	60/100	200/1000	260/1100

Let $Y = 1$ if a patient died and $Y = 0$ otherwise. Let X denote severity, taking the values serious and less serious.

- Compute $\bar{Y}$ for each mode of transport within each severity group. State which mode has the lower death rate among serious accidents, and which has the lower death rate among less serious accidents.
- Compute $\bar{Y}$ for each mode of transport across all its patients. State which mode has the lower death rate overall.
- Your answers to (a) and (b) disagree. Before explaining why, write each of the two pooled rates in (b) as a weighted average of that mode's two group rates from (a). Give the four weights as fractions, and say what each weight is the proportion of.
- The two modes have different weights. Name the feature of the transport system that produces that difference, using the sentence in the problem about when severity is assessed.
- Lecture wrote a pooled mean as $\bar{Y} = \sum_x \bar{Y}_{X=x} \hat{p}(x)$. Using that expression, state the two things that can differ between two groups of patients with different pooled rates. Say which one differs here.
- A newspaper reports the two numbers from (b) under the headline "Helicopter transport saves lives." State the question that comparison actually answers, and the question a reader takes it to answer.
- This problem demonstrates a paradox that was discussed in Week 1. Name it. The Week 1 example concerned income and vote choice measured on individuals and on states. State what that example and this one have in common, in terms of X , Y , and the weights in (c).

Part C — Expected utility

Question 3: The utility of voting

Lecture put a decision model to a pedestrian choosing between jaywalking and waiting. Put the same model to a citizen choosing between voting and abstaining on election day.

A citizen prefers one of the two candidates. Voting costs them $C = 5$, paid with certainty: the travel, the queue, the time off work. The vote changes the outcome of the election only if it is decisive, which occurs with probability p . If the vote is decisive, the citizen's preferred candidate takes office instead of the other one, which they value at $B = 500$. Abstaining costs nothing and changes nothing. Take $p = 1/50,000$.

- Let U_{vote} be the citizen's utility from voting. Write it as a random variable: give the two values it takes and the probability of each. Do the same for U_{abstain} .
- Compute $\mathbb{E}[U_{\text{vote}}]$ and $\mathbb{E}[U_{\text{abstain}}]$. State which action the model predicts.
- Rewrite $\mathbb{E}[U_{\text{vote}}]$ and $\mathbb{E}[U_{\text{abstain}}]$ in the parameters p , B and C . Give the condition on those three quantities under which the model predicts voting.

- d. Solve that condition with equality for p . Report the value of p at which the citizen is indifferent, given the B and C above. One vote in a national electorate is decisive with probability on the order of 10^{-7} . State what the model predicts for such an electorate.
- e. The model in (c) predicts near-universal abstention, and turnout in national elections is not near zero. One repair adds a term D , the utility of the act of voting itself, received whether or not the vote is decisive:

$$\mathbb{E}[U_{\text{vote}}] = pB - C + D.$$

Give the condition on D under which the model predicts voting, and compute the threshold value of D using the numbers above.

- f. Each of p , B , C and D differs across citizens. State in one sentence what a model of this form predicts about turnout across a population.
- g. Lecture gave three properties of expectation: $\mathbb{E}[c] = c$; $\mathbb{E}[cX] = c\mathbb{E}[X]$; and $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$. Using only these, show that $\mathbb{E}[a + bU] = a + b\mathbb{E}[U]$ for any constants a and b . Write one step per line. Name the property each step uses.
- h. Suppose every citizen's utility is measured on a new scale, $U' = a + bU$ with $b > 0$, applied to both actions. Using the result in (g), show that the action the model predicts in (c) is unchanged. Then state why adding a constant to the utility of voting alone is a different operation, and name the term in (e) that performs it.

Part D — Indicator variables

Question 4: Unseen vignettes

A survey experiment has $k = 10$ versions of a vignette. Each of $n = 20$ respondents is assigned one version, independently and uniformly at random.

- a. For version j , define $I_j = 1$ if no respondent sees version j , and $I_j = 0$ otherwise. Find $\mathbb{P}(I_j = 1)$ and $\mathbb{E}[I_j]$.
- b. Let U be the number of versions that no respondent sees. Write U as a sum of the indicators in (a), and find $\mathbb{E}[U]$.
- c. Are I_1 and I_2 independent? Explain why your answer to (b) does not depend on the answer.
- d. For each pair of respondents, define an indicator equal to 1 if the two saw the same version. Find the expected number of pairs who saw the same version.
- e. Simulate the assignment 10,000 times in R. Compare the mean of U with your answer to (b). Report the share of simulated experiments in which at least one version goes unseen.

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# sample(1:10, 20, replace = TRUE) assigns 20 respondents to 10 versions
# replicate() repeats an expression; unique() lists the versions that appear
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Part E — Variance and covariance

Question 5: Contact and turnout

A campaign's field team contacts some registered voters before an election. For a registered voter drawn at random, let $C = 1$ if the voter was contacted and $C = 0$ otherwise. Let $T = 1$ if the voter turned out and $T = 0$ otherwise. The joint distribution is below. (The probabilities are stipulated for the exercise.)

	$T = 0$	$T = 1$
$C = 0$	.30	.40
$C = 1$	.05	.25

- Find $\mathbb{E}[C]$. Find $\mathbb{V}[C]$ from the definition $\mathbb{V}[X] = \sum_i p_i (x_i - \mu)^2$. Check your answer against the Bernoulli variance from lecture.
- Find $\mathbb{V}[T]$ using the computational identity $\mathbb{V}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$. Explain why $\mathbb{E}[T^2] = \mathbb{E}[T]$ for any indicator T .
- Show that $\text{Cov}(C, T) = \mathbb{E}[CT] - \mathbb{E}[C] \mathbb{E}[T]$, following the steps lecture used for the computational identity. Then compute $\text{Cov}(C, T)$. Campaigns choose whom to contact. Name one way they choose that would produce this covariance even if contact had no effect on turnout.
- Let $K = C + T$. Find $\mathbb{V}[K]$ using the variance of a sum. Then find the distribution of K directly and confirm your answer. Report the value of $\mathbb{V}[K]$ you would get by wrongly treating C and T as independent.
- A survey records a respondent's self-placement as $X = -1$ (left), 0 (centre) or 1 (right), each with probability $1/3$. Let $Y = X^2$. Y equals 1 if the respondent takes a side and 0 if they place themselves at the centre. Show that $\text{Cov}(X, Y) = 0$. Then show that X and Y are not independent by finding values x and y with $\mathbb{P}(X = x, Y = y) \neq \mathbb{P}(X = x) \mathbb{P}(Y = y)$.